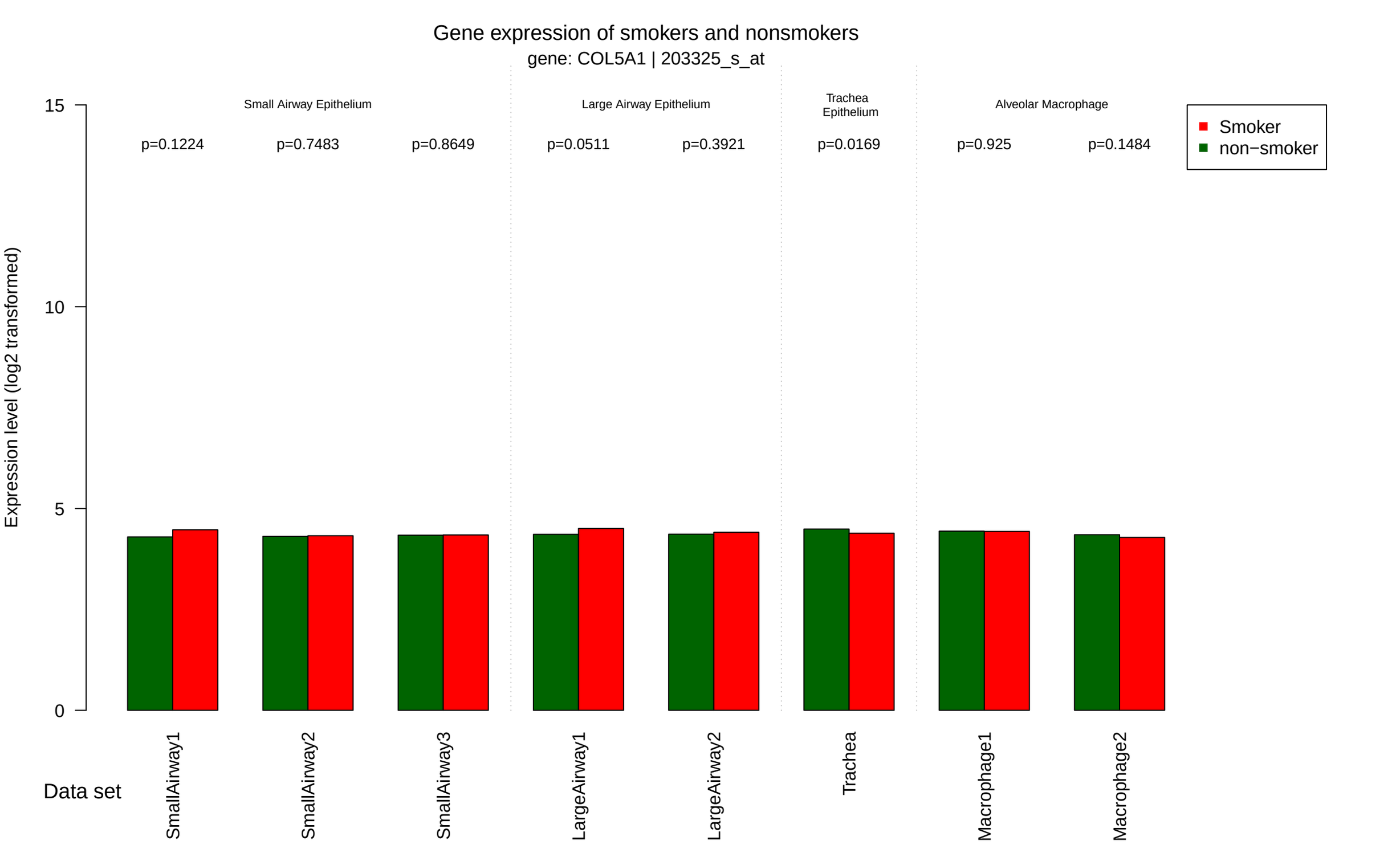
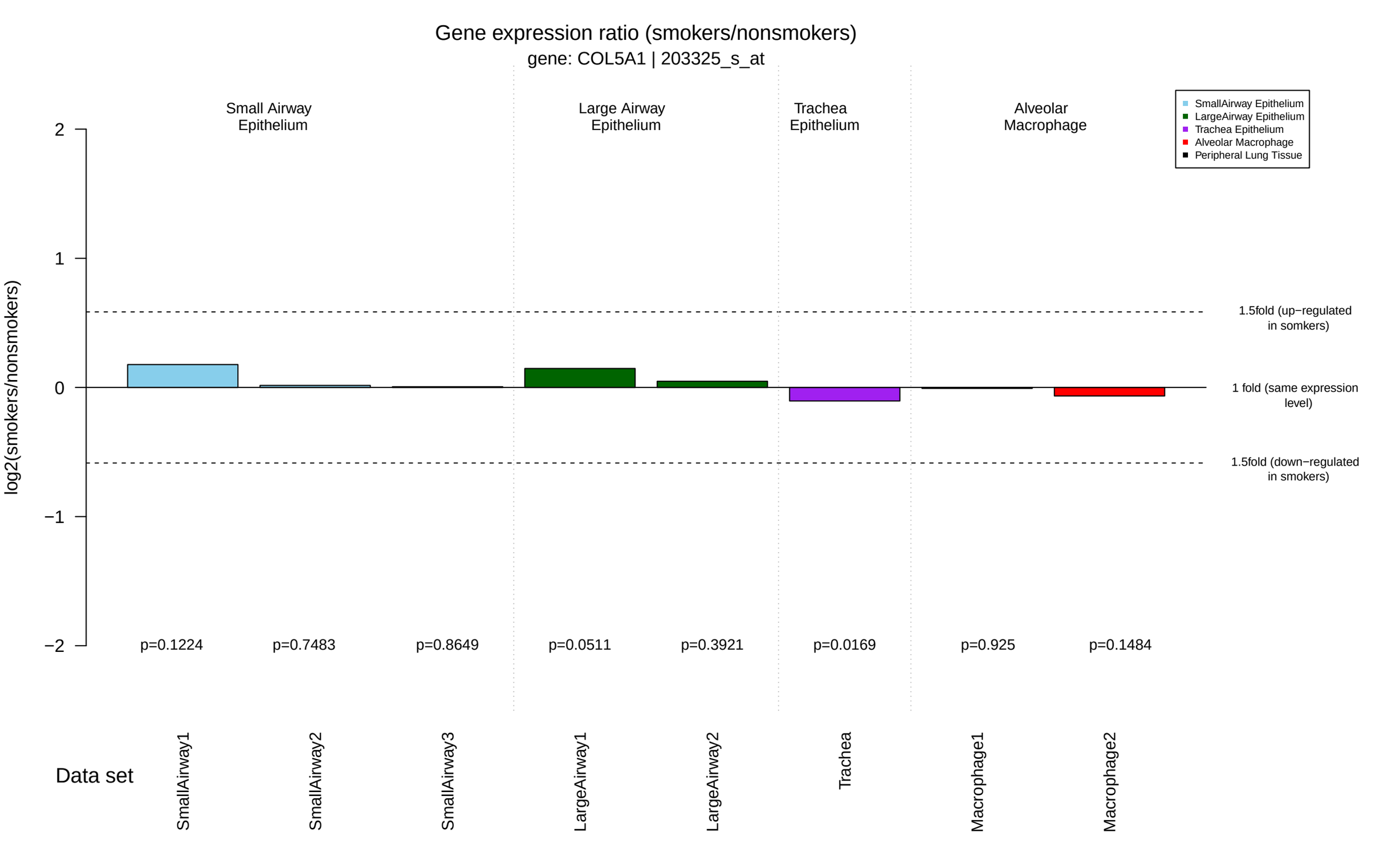
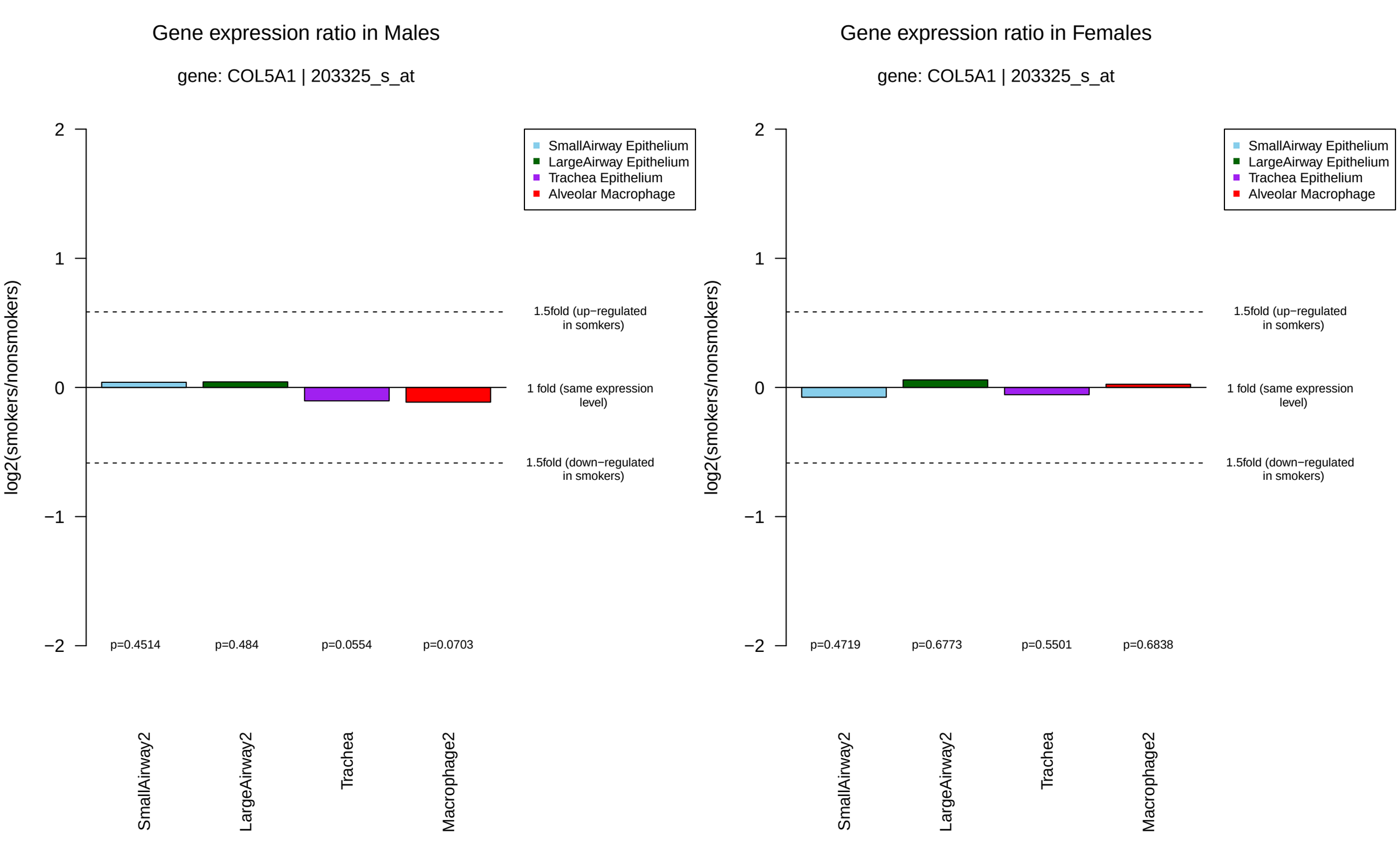
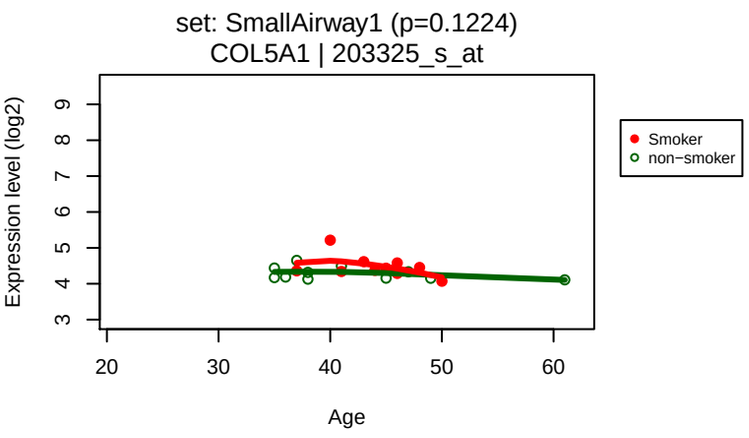
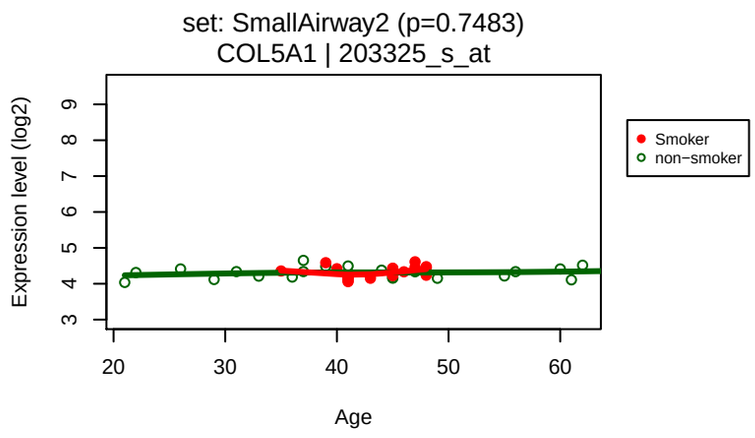
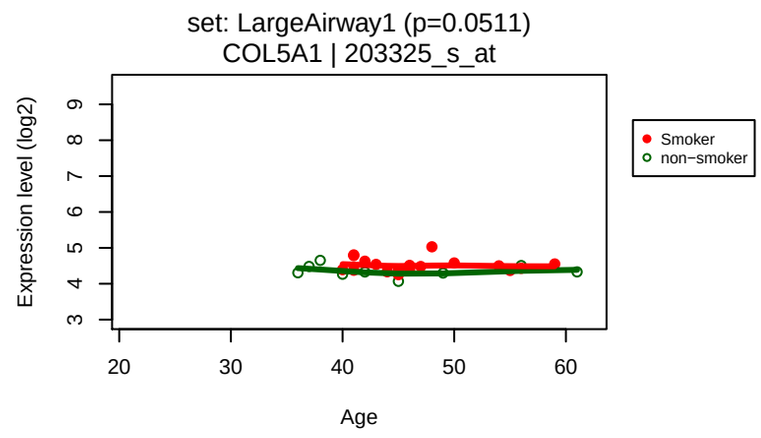

Gene expression of smokers and nonsmokers
gene: COL5A1 | 203325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1224
p=0.7483
p=0.8649
p=0.0511
p=0.3921
p=0.0169
p=0.925
p=0.1484
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

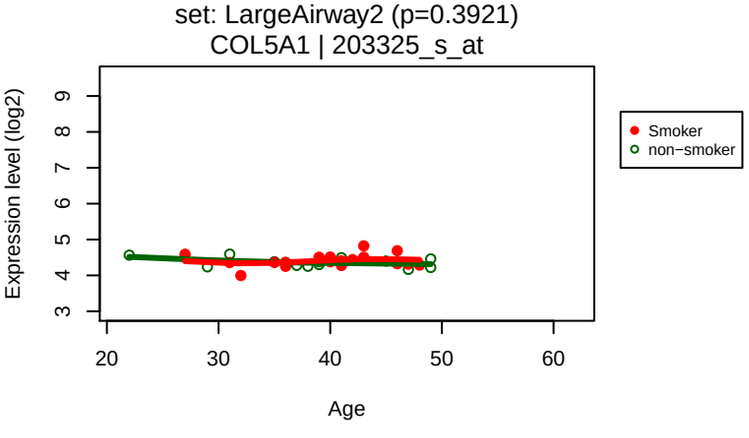

Gene expression ratio (smokers/nonsmokers)
gene: COL5A1 | 203325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1224
p=0.7483
p=0.8649
p=0.0511
p=0.3921
p=0.0169
p=0.925
p=0.1484
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

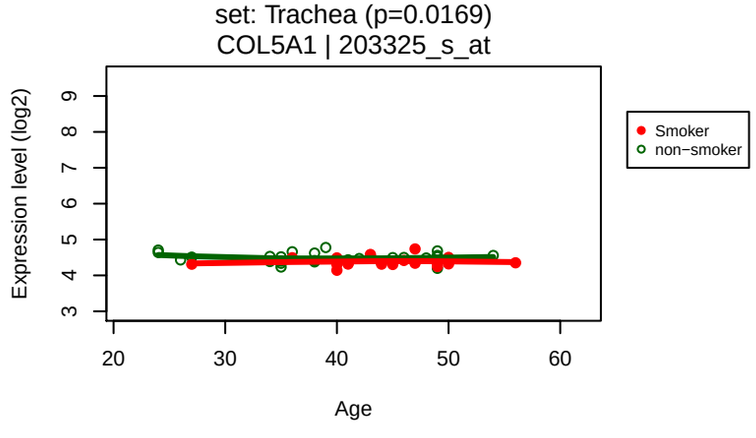

Gene expression ratio in Males
gene: COL5A1 | 203325_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4514
p=0.484
p=0.0554
p=0.0703
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL5A1 | 203325_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4719
p=0.6773
p=0.5501
p=0.6838
SmallAirway2
LargeAirway2
Trachea
Macrophage2

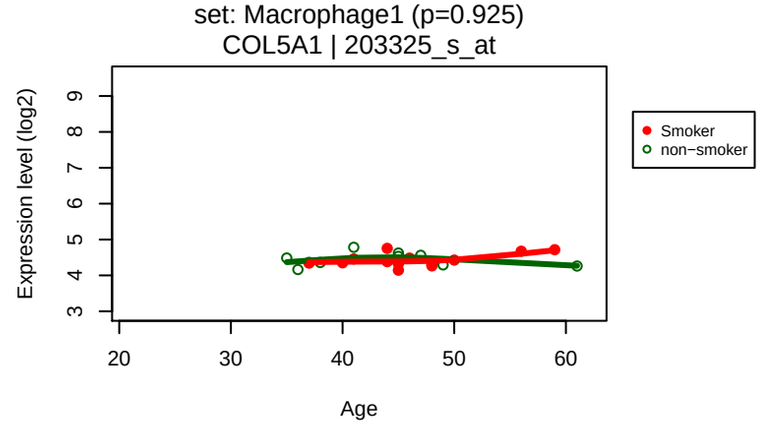

set: SmallAirway1 (p=0.1224)
COL5A1 | 203325_s_at
Expression level (log2)
Age
Smoker
non−smoker

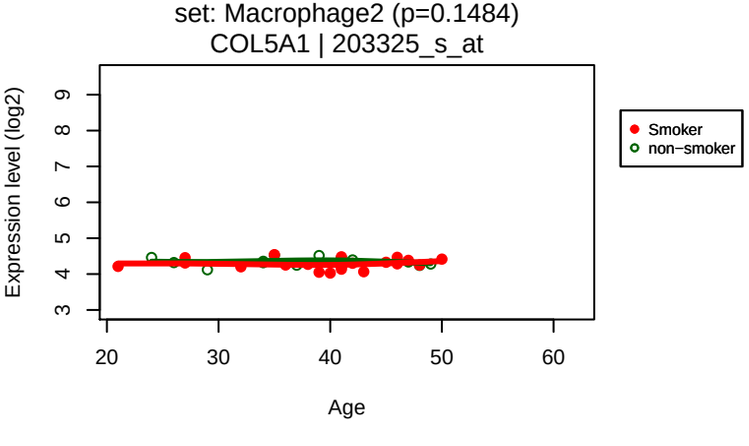

set: SmallAirway2 (p=0.7483)
COL5A1 | 203325_s_at
Expression level (log2)
Age
Smoker
non−smoker

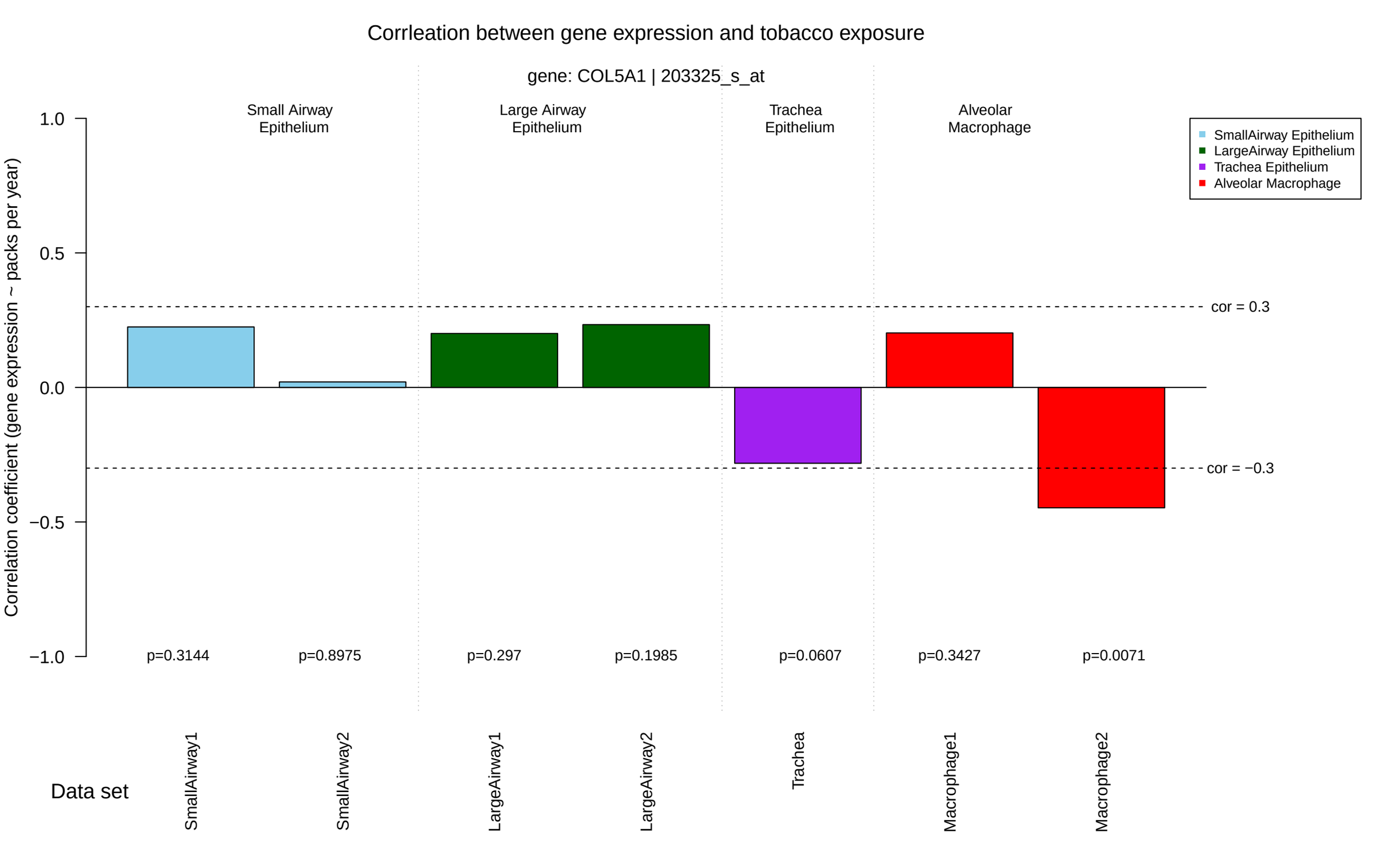

set: LargeAirway1 (p=0.0511)
COL5A1 | 203325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3921)
COL5A1 | 203325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0169)
COL5A1 | 203325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.925)
COL5A1 | 203325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1484)
COL5A1 | 203325_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL5A1 | 203325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3144
p=0.8975
p=0.297
p=0.1985
p=0.0607
p=0.3427
p=0.0071
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2